**This assignment will not be graded and is only for practice.**

## Multiple Choice Questions (MCQ):

1) Which of the following statements are correct? ***1 point***

- [ ] There exists a linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ such that $T(2,3) = (1,2)$, $T(1,-1) = (1,-1)$, and $T(4,1) = (1,0)$.

- [ ] If there is a linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ such that $T(2,3) = (1,2),$ and $T(1,-1) = (1,-1)$, then $T(x,y) = \frac{1}{5}(4x - y, -x + 4y)$.

- [ ] Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space $V$. Consider a linear transformation $T : V \to V$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. The matrix representation of $T$, with respect to $\beta$ and $\gamma$ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$

- [ ] Let $\{v_1, v_2, v_3\}$ be a basis of a vector space $V$ and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space $W$. Consider a linear transformation $T : V \to W$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. Then, the rank of $T$ is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If there is a linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ such that $T(2,3) = (1,2),$ and $T(1,-1) = (1,-1)$, then $T(x,y) = \frac{1}{5}(4x - y, -x + 4y)$.
Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space $V$. Consider a linear transformation $T : V \to V$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. The matrix representation of $T$, with respect to $\beta$ and $\gamma$ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$
Let $\{v_1, v_2, v_3\}$ be a basis of a vector space $V$ and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space $W$. Consider a linear transformation $T : V \to W$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. Then, the rank of $T$ is 3.

2) Consider the following set $S = \{(x,y) \mid x + y = 1, x, y \in \mathrm{R}\}$. In column A the matrix representation of some linear transformations are given with respect to the standard ordered bases. Match the entries in column A with the set $T(S)$ given in column B and the ***1 point***

geometric representations of $S$ and $T(S)$ in column C.

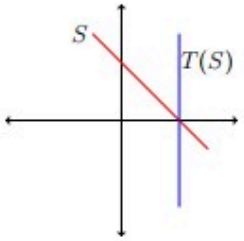
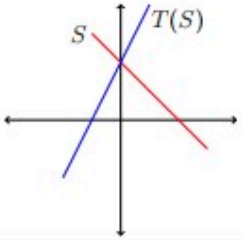
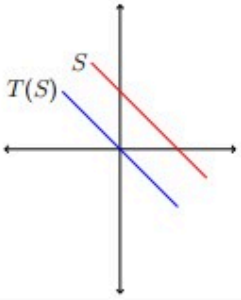
| | Matrix form of linear transformation (Column A) | | Image of the given set (Column B) | | Geometric representations (Column C) |
|---|---|---|---|---|---|
| i) | $T:\mathbb{R}^2 \to \mathbb{R}^2$ $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ | a) | $T(S) = \{(x,y) \mid x = 1, y \in \mathbb{R}\}$ | 1) |   |
| ii) | $T:\mathbb{R}^2 \to \mathbb{R}^2$ $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$ | b) | $T(S) = \{(x,y) \mid x + y = 0, x, y \in \mathbb{R}\}$ | 2) |   |
| iii) | $T:\mathbb{R}^2 \to \mathbb{R}^2$ $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ | c) | $T(S) = \{(x,y) \mid 2x - y = -1, x, y \in \mathbb{R}\}$ | 3) |   |

Table: M2W6P1

- [ ] i → b → 2.
- [ ] ii → c → 1.
- [ ] ii → c → 2.
- [ ] iii → a → 3.

- [ ] i → b → 3.
- [ ] iii → a → 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

ii → c → 2.
i → b → 3.
iii → a → 1.

**Multiple Select Questions (MSQ):**

3) Consider the following two groups. In Group 1, there are 3 functions from one vector space to another, and in Group 2, there are three properties of functions. All the vector spaces have usual addition and scalar multiplication. ***1 point***

**Group-1**

**P:** $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ **such that** $T(x,y) = \begin{cases} (\frac{x^2}{y}, y) & \text{if } y = 0 \\ (x, 0) & \text{otherwise} \end{cases}$

**Q:** $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$ **such that** $T(x,y) = (x, xy)$

**R:** $T : \mathrm{R}^3 \rightarrow \mathrm{R}$ **such that** $T(x,y,z) = \frac{x+y+z}{5}$

**Group-2**

1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$

2: $T(cv) = cT(v)$ for all $v \in V$ and $c \in \mathrm{R}$.

3: $T$ is a linear transformation

Choose the set of correct options.

- [ ] P satisfies 1 but not 2.
- [ ] P satisfies 2 but not 1
- [ ] Q satisfies 1
- [ ] Q satisfies 3
- [ ] P satisfies 3
- [ ] R satisfies 2
- [ ] R satisfies 3
- [ ] R satisfies 1 but not 2

No, the answer is incorrect.

Score: 0

Accepted Answers:

P satisfies 2 but not 1

R satisfies 2

R satisfies 3

## Numerical Answer Type (NAT):

4) Suppose the matrix representation of a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ with respect to the ordered bases $\beta = \{(1,0,0),(1,1,0),(1,1,1)\}$ for the domain and $\gamma = \{(1,1,1),(0,1,1),(0,0,1)\}$ for the range, is $I_{3\times 3}$, i.e., the identity matrix of order 3. Let $A$ be the matrix representation of the linear transformation $T$ with respect to standard ordered basis of $\mathrm{R}^3$ for both domain and range. Then the determinant of $A$ is

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

***1 point***

5) Let $T$ and $S$ be two linear transformations from $\mathrm{R}^3$ to $\mathrm{R}^3$, which are defined as follows:

$$T : \mathrm{R}^3 \to \mathrm{R}^3 \qquad S : \mathrm{R}^3 \to \mathrm{R}^3$$

$$T(x,y,z) = (x-y, y-z, z-x) \qquad S(x,y,z) = (x+y, y+z, z+x)$$

Let $S+T$ be defined to be the linear transformation as follows:

$(S+T) : \mathrm{R}^3 \to \mathrm{R}^3$

$(S+T)(x,y,z) = S(x,y,z) + T(x,y,z)$

Let $C$ be the matrix representation of $S+T$ with respect to the standard ordered bases of $\mathrm{R}^3$. If $C = nI$, where $I$ denotes the identity matrix of order 3, then what will be the value of $n$?

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

6) Consider a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ defined as $T(x, y, z) = (x + 5y, z - 3x, y + 6z)$. Which of the following options are true about $T$? ***1 point***

- [ ] Nullity of $T$ is $0$.
- [ ] A basis for the range of $T$ is $\{(1, 5, 0), (0, 1, 6)\}$.
- [ ] A basis for the range of $T$ is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
- [ ] Rank of $T$ is $2$.
- [ ] The matrix representation of $T$ with respect to the standard ordered bases of $\mathrm{R}^3$ is $\begin{bmatrix} 1 & -3 & 0 \\ 5 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.
- [ ] The matrix representation of $T$ with respect to the standard ordered bases of $\mathrm{R}^3$ is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.
- [ ] Nullity of $T$ is $1$.
- [ ] Rank of $T$ is $3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Nullity of $T$ is $0$.
A basis for the range of $T$ is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
The matrix representation of $T$ with respect to the standard ordered bases of $\mathrm{R}^3$ is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.
Rank of $T$ is $3$.

7) Let $V$ be a vector space with ordered basis $\alpha = \{v_1, v_2, \ldots, v_{10}\}$. Consider a linear transformation $T : V \rightarrow V$ such that $T(v_1) = v_1$ and $T(v_i) = v_i - v_{i-1}$, where $i > 1$. Let $A$ be the matrix representation of the linear transformation $T$, with respect to $\alpha$ for both the domain and codomain. what is the trace of $A$?

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

***1 point***

**Comprehension Type Question:**

For each video on YouTube, consider the vector (Number of comments in thousands, Number of shares in thousands, Number of views in thousands) and consider the subset $S$ of $\mathrm{R}^3$ consisting of these vectors. It is known that the ranking of a YouTube video can be thought of as the restriction of a linear transformation $T : \mathrm{R}^3 \rightarrow \mathrm{R}$ to $S$ where $\mathrm{R}^3$ comes with usual addition and scalar multiplication. (assume ranking 0 (zero) is possible and in practice negative rankings can be ignored).
The following data in Table M2W6P2 shows the correlation between comments, shares, and views - with YouTube rankings of a video.

| Number of comments (in thousands) | Number of shares (in thousands) | Number of views (in thousands) | Ranking of a video |
|---|---|---|---|
| 3 | 10 | 8 | 3 |
| 4 | 15 | 10 | 2 |
| 5 | 20 | 15 | 1 |

Table: M2W6P2

Use the above information to answer questions 8, 9 and 10.

8) Which of the following sets of vectors in $S$ have the property that the YouTube video corresponding to every linear combination of that set which is in $S$ has ranking $0$? ***1 point***

- [ ] $\{(1, 1, 0), (0, 0, 1)\}$
- [ ] $\{(6, 25, 0), (0, 0, 25)\}$
- [ ] $\{(25, 6, 0), (0, 0, 25)\}$
- [ ] $\{(12, 50, 0), (0, 0, 1)\}$
- [ ] $\{(24, 100, 0), (0, 0, 25)\}$
- [ ] $\{(6, 0, 0), (0, 25, 0), (0, 0, 1)\}$
- [ ] $\{(25, 0, 0), (0, 6, 0), (0, 0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(6, 25, 0), (0, 0, 25)\}$
$\{(12, 50, 0), (0, 0, 1)\}$
$\{(24, 100, 0), (0, 0, 25)\}$

9) Which of the following sets of vectors in $S$ have the property that the YouTube video corresponding to every linear combination of that set which is in $S$ has ranking a multiple of $15$? ***1 point***

- [ ] $\{(3, 0, 0), (0, 0, 1)\}$
- [ ] $\{(3, 0, 1)\}$
- [ ] $\{(3, 0, 1), (0, 0, 1)\}$
- [ ] $\{(0, 0, 1)\}$
- [ ] $\{(9, 25, 0), (0, 0, 1)\}$
- [ ] $\{(25, 9, 0), (0, 0, 1)\}$
- [ ]

$\{(9,0,0),(0,25,0),(0,0,1)\}$

- [ ] $\{(25,0,0),(0,9,0),(0,0,1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(3,0,0),(0,0,1)\}$
$\{(3,0,1)\}$
$\{(3,0,1),(0,0,1)\}$
$\{(0,0,1)\}$
$\{(9,25,0),(0,0,1)\}$
$\{(9,0,0),(0,25,0),(0,0,1)\}$

10) Suppose that for a YouTube video there are 5 (in thousands) shares. What must be the number of comments (in thousands) so that the ranking of the video will be 4? [Note: Suppose your answer is 6000, then you have to enter 6 as the answer.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

Your score is: 0/10